

HICKS BAY, New Zealand

WMO: 931960

Lat: 37.5611S

Lon: 178.3144E

Elev: 46

StdP: 100.77

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.9	6.0	-0.3	3.7	6.9	1.1	4.1	7.7	18.5	13.3	17.3	12.7	4.1	220	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.2	23.4	19.8	22.6	19.2	21.9	18.8	20.9	22.2	20.2	21.7	19.7	21.2	7.1	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.4	15.2	21.6	19.7	14.5	21.2	19.1	13.9	20.6	60.7	22.2	58.5	21.8	56.4	21.2	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.6	13.1	DB	2.8	24.9	0.6	1.3	2.4	25.8	2.0	26.5	1.7	27.3	1.2	28.2	(4)
(5)			WB	1.2	21.7	0.8	0.8	0.6	22.3	0.1	22.8	-0.3	23.2	-0.9	23.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	18.5	19.1	18.0	16.2	14.2	12.0	11.4	11.5	12.5	13.7	15.0	17.1	(6)	
	DBStd	3.27	1.72	1.59	1.78	2.04	2.16	2.16	1.95	1.77	1.73	1.71	1.82	1.79	(7)	
	HDD10.0	24	0	0	0	0	1	6	8	7	2	1	0	0	(8)	
	HDD18.3	1331	19	9	27	67	128	189	215	211	174	145	102	45	(9)	
	CDD10.0	1822	264	256	249	187	131	67	52	54	78	114	149	221	(10)	
	CDD18.3	86	24	32	18	4	1	0	0	0	0	0	1	7	(11)	
	CDH23.3	23	7	13	3	1	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	6.8	6.6	6.4	6.4	6.3	6.6	6.8	6.9	6.7	7.2	7.3	7.2	6.5	(14)	
(15) Precipitation	PrecAvg	1802	111	114	134	148	170	202	204	189	153	124	102	125	(15)	
	PrecMax	2194	299	311	374	330	341	327	379	326	258	271	204	347	(16)	
	PrecMin	1437	17	13	39	80	39	85	53	126	92	50	35	33	(17)	
	PrecStd	217	65	82	73	59	78	68	71	54	40	56	42	78	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.2	24.6	23.6	21.9	20.1	17.7	16.7	16.5	17.9	18.7	21.1	22.5	(19)	
		MCWB	20.3	20.7	19.7	18.6	16.8	15.8	14.7	13.9	14.4	15.3	17.7	19.1	(20)	
	2%	DB	23.2	23.6	22.6	20.8	19.0	16.9	15.9	15.6	16.7	17.9	19.8	21.6	(21)	
		MCWB	19.7	20.0	19.1	18.1	16.9	15.2	14.0	13.4	14.1	15.0	16.8	18.6	(22)	
	5%	DB	22.3	22.8	21.8	20.1	18.2	16.3	15.2	15.1	16.0	17.2	18.8	20.8	(23)	
		MCWB	19.2	19.4	18.6	17.5	16.3	14.5	13.2	13.1	13.7	14.5	16.0	18.0	(24)	
	10%	DB	21.6	22.0	21.1	19.4	17.7	15.6	14.7	14.6	15.4	16.6	18.0	20.0	(25)	
		MCWB	18.7	18.9	18.3	17.0	15.7	13.8	12.6	12.5	13.3	14.2	15.5	17.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.5	21.8	21.0	20.3	18.6	16.7	15.7	15.2	15.7	16.6	18.5	20.2	(27)	
		MCDB	23.1	23.1	22.2	21.2	19.0	17.1	16.3	15.6	16.5	17.5	20.1	21.3	(28)	
	2%	WB	20.7	21.1	20.1	19.1	17.7	16.0	14.7	14.5	15.1	15.9	17.5	19.4	(29)	
		MCDB	22.2	22.4	21.4	20.0	18.3	16.6	15.4	15.1	15.9	17.0	18.9	20.5	(30)	
	5%	WB	20.0	20.5	19.5	18.4	17.0	15.2	13.9	13.8	14.5	15.3	16.7	18.8	(31)	
		MCDB	21.5	21.9	21.1	19.5	17.9	16.0	14.9	14.7	15.5	16.5	18.1	20.0	(32)	
	10%	WB	19.3	19.8	18.8	17.6	16.1	14.2	13.0	13.0	13.9	14.7	16.1	18.2	(33)	
		MCDB	20.9	21.4	20.6	19.0	17.5	15.4	14.4	14.2	15.0	16.1	17.6	19.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	5.2	5.4	5.4	5.3	5.3	5.1	5.4	5.2	5.3	5.2	5.0	(35)	
		MCDBR	5.8	5.8	6.0	5.5	5.2	4.7	4.9	5.3	5.3	5.6	5.8	5.5	(36)	
	5% WB	MCWBR	4.1	3.9	4.1	4.0	4.0	4.2	4.1	4.3	4.2	4.4	4.5	4.1	(37)	
		MCWBR	4.9	4.6	4.8	4.1	3.6	3.8	3.9	4.1	4.2	4.7	5.1	4.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.350	0.346	0.332	0.318	0.307	0.307	0.303	0.305	0.328	0.340	0.339	0.346	(40)	
		taud	2.568	2.599	2.634	2.661	2.661	2.631	2.644	2.643	2.556	2.531	2.551	2.577	(41)	
	Ebn at Noon		984	964	939	894	846	813	840	895	927	959	987	991	(42)	
		Edh at Noon	105	98	88	75	65	63	65	75	93	104	107	106	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.86	5.78	4.69	3.40	2.43	1.92	2.10	2.98	4.13	5.54	6.30	6.65	(44)	
	RadStd		0.40	0.36	0.35	0.20	0.14	0.13	0.12	0.17	0.34	0.33	0.37	0.45	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.35	N/A	N/A	+0.60	+0.71	+0.66	N/A	N/A	N/A	+29
(47) Regional (2 neighbors)		+0.32	+0.33	+0.43	+0.45	N/A	N/A	-11	-80	+106	+29

Nomenclature: See separate page